

Review

Informing deep neural networks by multiscale principles of neuromodulatory systems

Jie Mei,^{1,7,*} Eilif Muller,^{2,3,4} and Srikanth Ramaswamy^{5,6,*}

Our brains have evolved the ability to configure and adapt their processing states to match the unique challenges of acting and learning in diverse environments and behavioral contexts. In biological nervous systems, such state specification and adaptation arise in part from neuromodulators, including acetylcholine, noradrenaline, serotonin, and dopamine, whose diffuse release fine-tunes neuronal and synaptic dynamics and plasticity to complement the behavioral context in real-time. Despite the demonstrated effectiveness of deep neural networks for specific tasks, they remain relatively inflexible at generalizing across tasks or adapting to ever-changing behavioral demands. In this article, we provide an overview of neuromodulatory systems and their relationship to emerging pertinent principles in deep neural networks. We further outline opportunities for the integration of neuromodulatory principles into deep neural networks, towards endowing artificial intelligence with a key ingredient underlying the flexibility and learning capability of biological systems.

Biological neuromodulation and deep neural networks (DNNs)

Neuromodulators are chemical messengers that diffusely target neurons and synapses in specific brain regions. Distinct groups of neurons located in the midbrain, hindbrain, or forebrain areas release neuromodulators such as acetylcholine (ACh), noradrenaline (NA), serotonin (5-HT), and dopamine (DA), with region-specific yet partly overlapping spatial innervation patterns (Figure 1A) [1]. Neuromodulators are typically released through presynaptic swellings (varicosities) on *en passant* axon collaterals (see Glossary) and diffuse in the local volume to target all structures in their vicinity (known as paracrine signaling), with differential, cell type-specific effects [2]. Neuromodulators bind to an assortment of receptors distributed across neurons (including their axons, dendrites, and synapses) as well as other cell types and operate with diverse temporal dynamics to control ion channel kinetics and cellular excitability, fine-tune synaptic transmission, gate plasticity, and reconfigure network dynamics [3]. For example, in biological neural circuits, such as central pattern generators (CPGs), neuromodulators configure the parameters of network components: neurons and synapses [4]. Blocking all modulatory inputs in CPGs and applying individual neuromodulators, such as 5-HT and DA, can induce different spiking patterns in the same class of neurons [5]. Therefore, specific neuromodulators can regulate neuronal activity in CPGs to reconfigure the same network, resulting in different functional outputs. These general design principles of neuromodulation are broadly conserved in nervous systems across animal species.

This review aims to lay out the multiscale organizing features of biological neuromodulatory systems in learning and cognition and examine their diverse functional settings that could inspire new approaches to inform and evaluate deep neural networks (DNNs). We also briefly discuss some of the ongoing research incorporating biologically plausible neuromodulatory mechanisms into functional DNN models.

Highlights

Neuromodulatory systems in biological brains play an important role in learning and adaptive behavior.

Different neuromodulators operate on a spectrum of spatio-temporal scales in tandem and opposition to reconfigure functions of local neural networks and to regulate global cognitive states.

Although neuromodulators are important for general cognition, their phenomenology is yet to be fully realized in deep neural networks (DNNs) and has so far been mostly limited to implementations of reinforcement learning models.

We highlight the organizing principles of neuromodulators in adaptive cognition, explore their diverse functional settings to inspire new approaches of DNNs, and offer suggestions on how these approaches could be evaluated.

¹Department of Anatomy, Université du Québec à Trois-Rivières, Trois-Rivières, Canada

²Department of Neurosciences, Université de Montréal, Montréal, Canada

³CHU Sainte-Justine Research Center, Montréal, Canada

⁴Quebec Artificial Intelligence Institute (Mila), Montréal, Canada

⁵Institute of Physiology, University of Bern, Switzerland

⁶Biosciences Institute, Newcastle University, UK

⁷Current address: The Brain and Mind Institute, University of Western Ontario, London, ON, Canada; Department of Computer Science, University of Western Ontario, London, ON, Canada

*Correspondence: jmei47@uwo.ca (J. Mei) and srikanth.ramaswamy@newcastle.ac.uk (S. Ramaswamy).



Neuromodulatory systems and their functions

In the mammalian brain, long-range axonal projections from the neuromodulatory nuclei release ACh, NA, 5-HT, and DA into brain regions, including the hippocampus [6], neocortex [7,8], and striatum [9], and regulate short- and long-term cellular excitability and synaptic plasticity (Figure 2) to shape decision making, learning, attention, memory, and motivation [10–15]. The concentration and spatio-temporal profile of the release of neuromodulators synergize with the responsiveness, dynamics, and localization of diverse receptors in postsynaptic targets. Combined with cell type-specific expression of these receptors, such interactions dictate the specific spatio-temporal patterns and timescales of neuromodulatory effects in distinct cell types, synapses, and networks [16]. For example, neuromodulation could enable learning by bridging the sub-millisecond precision of local learning mechanisms, such as spike timing-dependent plasticity (STDP), with the slower timescales of behavioral signals, including reward or aversive stimuli [17]. Specifically, neuromodulators such as ACh could prime synapses for the subsequent induction of plasticity on variable timescales (prospective neuromodulation of STDP) [18], or act with a delayed window of seconds to minutes, as observed with DA in the hippocampus (retrospective neuromodulation of STDP) [19]. Overall, the multiscale organization of neuromodulatory systems provides high-level contextual control of cognition and behavior. This contextual control operates on the one hand, by reshaping local cellular and synaptic physiology and, on the other hand, by modulating global network properties, which in the context of DNNs, could be considered as adjustment of hyperparameters, such as learning rate and network complexity. In this way, neuromodulatory processes may fine-tune hyperparameters to dynamically adapt learning strategies of DNNs (Figure 1B). In a way, neuromodulatory systems can be seen as 'matchmakers' between the environment and the brain, by inferring behavioral contexts and broadly adapting multiscale properties and parameters to gate learning and enable transitions between behavioral states such as sleep and wakefulness, or distraction and attention [20–22].

In some situations, neuromodulators can act in concert, but in other contexts, they compete to shape cognitive processes. For instance, coordinated interactions between ACh and 5-HT in the hippocampus, a brain region of key relevance for learning and memory, are important for consciousness and cognitive functioning [23,24]. In addition, these interactions bring about transitions between brain states such as distraction and attention or exploration and exploitation [25–27] and in governing goal-directed, task-driven cognitive behaviors such as working memory and action selection [28]. Conversely, in the prefrontal cortex, which is another crucial hub for learning and attention processing, ACh and 5-HT have competing effects on neurons located deep in layer 6, an emerging key player in attentional control. Opposing interactions between ACh and 5-HT result in an attentional 'tug-of-war' by enhancing and suppressing, respectively, patterns of action potential generation in the same neuron type [29]. Both ACh and 5-HT are actively released during wakefulness, with an increase in ACh levels during attention. Functionally, the opposing role of 5-HT as a brake to counter ACh modulation in the prefrontal cortex appears consistent with its role in modulating cognitive and behavioral flexibility [30,31]. This competing phenomenon could be relevant for abnormalities in attention that are implicated in neurological disorders associated with disrupted release of 5-HT.

Emerging evidence has also identified competitive, dynamic interactions between DA and 5-HT in the prefrontal cortex, amygdala, and the striatum. These interactions contribute to neural processes underlying reward-seeking behavior, reinforcement learning, and punishment avoidance [32]. For instance, an important association of the functional opponency between DA and 5-HT is in the context of behavioral inhibition in the face of predictions of aversive outcomes. It appears that DA influences vigor in the face of hunger [33], whereas 5-HT modulates quiescence in the face of satiety [34,35]. Moreover, 5-HT neurons are activated in the face of punishment [36–38], whereas DA neurons are associated with aversive paradigms [32].

Glossary

Activation function: a class of functions that transforms the weighted sum of an input into an output and is commonly used to introduce nonlinearity to neural networks to facilitate learning of complex patterns.

Axon collateral: a side branch of an axon that further splits into terminal branches. An axon may have multiple collaterals that target different structures.

Catastrophic forgetting: the loss and forgetting of knowledge of previously learned task(s) when a neural network is being exposed to information relevant to a new task.

Compartmental model neuron: an approach that defines the cellular morphology and membrane parameters of single neurons and kinetic models for individual compartments, to enable biologically plausible modeling of neuronal activities.

Deep neural network (DNN): a neural network with multiple hidden layers between the input and output layers.

Deep reinforcement learning: the use of deep neural networks and representations learned by deep neural networks to solve reinforcement learning problems.

Hebbian learning: a learning mechanism proposed by Donald Hebb describing how the connection between two neurons alters depending on their activity. Correlated activation leads to strengthened connectivity and separate activation leads to reduced connectivity.

Hyperparameter: a parameter used directly in the control of a model's learning process, which is manually set prior to the start of learning process.

Learning rate: a hyperparameter that defines the magnitude of parameter updates during model training.

Loss function: a function that computes the difference between the output of a model and the target value.

Microcircuit: an elementary processing module governed by connectivity and organization of neurons within certain brain regions, allowing for functional adaptation for specific tasks.

Network topology: a description of the relationship between individual neurons and the way they are interconnected to form a network that enables learning and other biological or cognitive functions.

Reward prediction error (RPE): the difference between received and predicted rewards.

The delicate interplay of ACh and DA signaling in the striatum is known to be crucial for maintaining motor control and the function of the basal ganglia [39] and is essential in the learning and execution of motor sequences [40]. Striatal microcircuitry is balanced by the so-called direct and indirect pathways, which ultimately (as a simplified summary) excite and inhibit the neocortex to control motor learning and movement [41]. ACh and DA antagonistically control the functions of both the direct and indirect pathways. Suppressing the activity of cholinergic cells reduces the release of ACh, resulting in a predominance of DA, which excites and mutes the direct and indirect pathways, respectively. Consequently, this leads to increased excitatory output from the striatum to the neocortex, which induces abnormal motor learning [41]. By contrast, the release of ACh from striatal cholinergic neurons is thought to inhibit and excite the direct and indirect pathways, respectively. Cholinergic cells, however, comprise only about 2% of all neurons in the striatum and their precise function in striatal circuitry is still not fully understood [42]. The synergistic and opposing functional roles of neuromodulators have led to hypotheses on how and why diverse brain regions interact collaboratively and competitively to achieve computational goals in decision making and learning (Box 1).

Neuromodulation, adaptive learning, and behavioral flexibility

In biological nervous systems, learning occurs through continuous observation and interaction with the surrounding environment [43,44]. Experimental studies using animal models have revealed several distinct features of biological neuromodulation that could help enable long-term, adaptive learning. For example, DA neurons in the midbrain ventral tegmental area are modulated by presentation of rewards or salient, nonrewarding stimuli. DA neurons display both tonic (sustained) and phasic (bursting) patterns of firing activity [45,46], which have been associated with different functions. The slower, tonic background activity of DA has been hypothesized to

Signaling: the transmission and propagation of signals for integration and transfer of information within and between neurons and subneuronal compartments.

Spike timing-dependent plasticity (STDP): a temporally asymmetric form of Hebbian learning where the precise timing of spikes of pre- and postsynaptic neurons significantly affects synaptic plasticity.

Synaptic plasticity: activity-dependent modification of: (i) the strength of synaptic transmission, (ii) size or number of synapses, or (iii) morphological structures containing synapses (e.g., dendrite spines). Synaptic plasticity can be broadly divided into short-term synaptic plasticity and long-term plasticity.

Volume transmission: long distance signaling that involves actions of neurotransmitters at a distance beyond the immediate region of their initial release sites, enabling a temporally slower and spatially broader mode of intercellular communication compared with synaptic transmission.

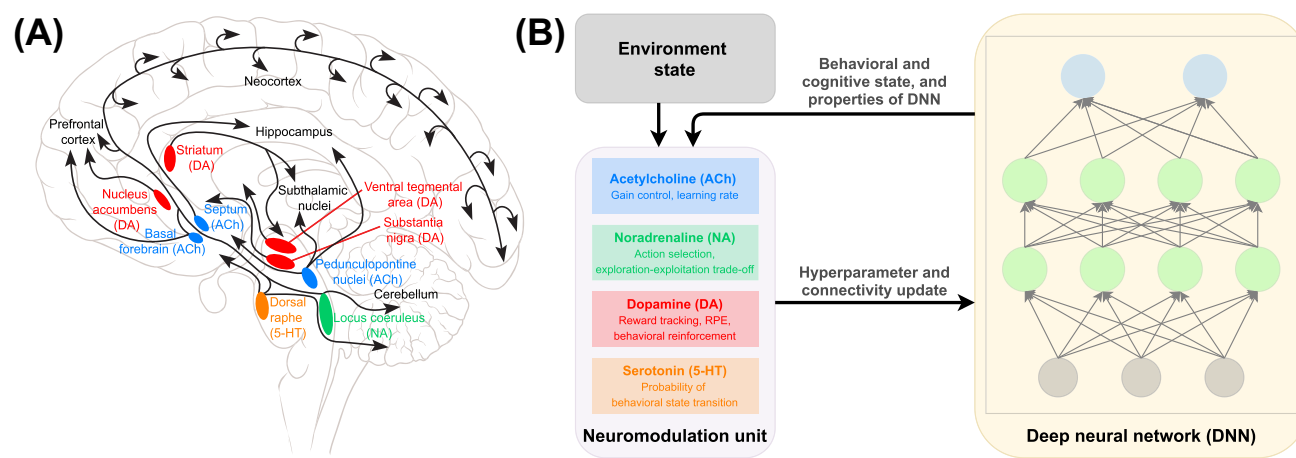


Figure 1. Biological neuromodulatory systems and neuro-inspired artificial neuromodulation. (A) Individual neuromodulatory systems originate from and target diverse brain regions, giving rise to diverse spatial and temporal functional effects. Dopaminergic neurons (red) in the substantia nigra and ventral tegmental area project to the striatum, the limbic system, and prefrontal cortex, regulating reward processing and motivation. Noradrenergic neurons (green) in the locus coeruleus project to cortical and subcortical regions and target the neocortex, hippocampus, and cerebellum, controlling processes such as arousal, attention, and behavioral adaptation. Cholinergic projections (blue) originate from neurons of the medial septum, nucleus basalis of Meynert, and pedunculopontine nucleus and innervate the thalamus, basal ganglia, hippocampus, and cortex. The cholinergic system contributes to attention and memory processes. Serotonergic neurons (orange) from the dorsal raphe nucleus innervate the neocortex, cerebellum, and limbic system, mediating memory, cognition, reward, cost assessment and physiological functions, including feeding and sleep. (B) Diagram of a hypothetical artificial neuromodulation unit to be integrated with a deep neural network (DNN). The neuromodulation unit in this example consists of four neuromodulators. This neuromodulation unit receives representations of current states of the environment, as well as behavioral and cognitive state (e.g., the chosen action, or the level of attention/alertness), and properties of the DNN (e.g., number of layers and learning rate). In turn, it reconfigures the DNN with updated hyperparameter values (e.g., learning rate). Abbreviations: ACh, acetylcholine; DA, dopamine; 5-HT, serotonin; NA, noradrenaline; RPE, reward prediction error.

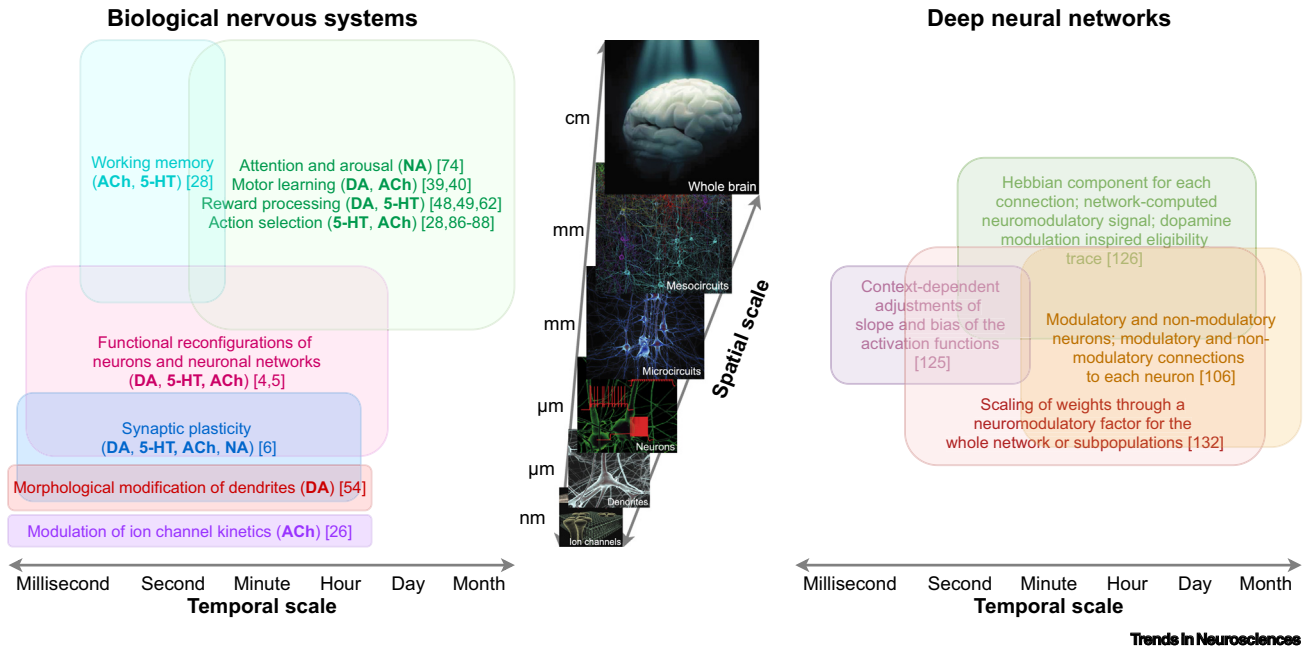


Figure 2. Multiscale functional effects of neuromodulation in biological organisms and neuromodulated deep neural networks. An overview of the spatial and temporal scales of (A) neuromodulated physiological and cognitive processes in the biological nervous systems, and (B) implementations of neuromodulation in deep neural networks (DNNs). Images for ion channels, dendrites, neurons, microcircuits, mesocircuits, and whole brain are credited to the Blue Brain Project, EPFL. Abbreviations: ACh, acetylcholine; DA, dopamine; 5-HT, serotonin; NA, noradrenaline. See [4–6,26,28,39,40,48,49,54,62,74,86–88,106,125,126,132].

Box 1. Neuromodulation-inspired DNN models of learning and action selection

Dopamine, serotonin, and the temporal difference (TD) error

Actor–critic architectures are widely used in reinforcement learning and are inspired partly by reinforcement learning mechanisms in the basal ganglia. In this architecture, an actor selects an action based on the future reward predicted by the critic. The critic learns to predict future rewards through dopamine-like reinforcement signals. Specifically, the actor–critic architecture uses the TD error, the difference between predicted values over consecutive time points to update value estimation, to improve the state of an agent [136]. In this conceptualization, the agent can be either an animal/human or a machine, which perceives its environment and takes actions autonomously in order to achieve learning goals.

Previous studies have suggested that the reward prediction error is signaled by the dopaminergic system [57,144]:

$$\delta_t = r_{t+1} + \gamma V(s_{t+1}) - V(s_t) \quad [I]$$

Where δ_t represents the TD error at time t , r_{t+1} is the observed reward at time $t + 1$, $V(s_{t+1})$ and $V(s_t)$ are the values for the agent of state s at time $t + 1$ and t , respectively, and γ is the discounting factor. It has been hypothesized that the direct and indirect inhibitory connections from the striatum to the substantia nigra pars compacta compute the value function V , while serotonergic levels may regulate reward discounting via cortico-basal ganglia loops [22].

Noradrenaline and exploration–exploitation trade-off

Noradrenergic neurons originate in the locus coeruleus (LC). The LC has been implicated in exploration–exploitation trade-offs [61]. Mathematically:

$$P(a_i|s) = \frac{\exp(\beta Q(s, a_i))}{\sum_{j=1}^n \exp(\beta Q(s, a_j))} \quad [II]$$

where Q is the value function given action a_i and state s , a_i represents an action among a total of n actions ($i = 1 \dots n$) and β is a parameter controlling the exploration–exploitation trade-off. Tonic activity in LC leads to a search for alternatives, while phasic LC activity helps optimize performance in a task [74]. LC receives input from the anterior cingulate cortex and orbitofrontal cortex, both of which regulate transitions between phasic and tonic network activities. A higher β means lower randomness in action selection.

regulate the level of motivation [47], while phasic activity has been suggested to signal the **reward prediction error (RPE)** (the discrepancy between expected and actual rewards) [48,49], a central element in reinforcement learning [50].

Consistent with this perspective, a recent study in primates suggests that a subset of DA neurons tracked changes in the magnitude of reward through tonic firing, while phasic activity was evoked in most DA neurons when learned associations between visual cues and a reward were violated [51]. Importantly, these results indicate that there is a partial overlap between the populations encoding the phasic versus tonic mechanisms and neurons may switch between these modes depending on behavioral context. Electrophysiological recordings in freely moving rodents have demonstrated that the phasic activity of midbrain DA neurons can affect cortical activity on rapid timescales, altering the firing rate of some regular- and fast-spiking neurons in a transient manner [52]. At slower time-scales, these phasic activities also promote the coupling of gamma and theta band brain oscillations, and increased gamma power in medial prefrontal cortex, which is associated with the reallocation of cognitive resources to support decision making [52,53]. Locally, transient DA release has been observed to induce enlargements of dendritic spines in striatal neurons over a time window ranging from 0.3 to 2 seconds [54], providing a cellular and synaptic basis for reinforcement plasticity [54–57].

Recent studies on anatomically segregated and functionally distinct 5-HT pathways have revealed an unexpected yet sophisticated role for 5-HT in governing behavior [58,59]. For example, in a process known as temporal discounting, where the subjective value of a reward decreases in magnitude when the given reward is delayed [22,60,61], 5-HT regulates impulsivity in decision making and controls the processing of rewards that occur at different times. In rodents with inhibited 5-HT activity, waiting for long-delayed rewards was impaired, while waiting for short-delayed rewards was unchanged, suggesting that the activation of serotonergic neurons in midbrain structures, such as the dorsal raphe, is necessary for waiting for long-delayed rewards [62]. The activity of 5-HT neurons has been hypothesized to promote patience and waiting by increasing reward confidence under high reward timing uncertainty [63]. Although models of reinforcement learning are often centered around DA signaling, recent studies have attempted to map the emerging role of dorsal raphe 5-HT neurons in modulating learning rates in rodents undertaking a reward-driven decision-making task [64]. The dorsal raphe comprises neuronal populations with heterogeneous neurochemical and anatomical properties [65]. Among these, approximately half release 5-HT and the remainder release other neuromodulators, including DA and a host of neuropeptides such as galanin and corticotropin-releasing factor [66]. The interactions of these diverse neuronal populations in the dorsal raphe is still unknown. A systematic dissection of neuronal properties in the dorsal raphe and how they shape decision making could yield insights to inform future models of reinforcement learning [67].

Context-based state transitions and information processing

The brain implements various strategies to learn and adapt. Some of the learning strategies that have been often discussed in association with specific brain structures include reinforcement learning in the cortico-basal ganglia loop [68], supervised learning in the cerebellum [69], episodic memory and incremental learning in the hippocampal region [70], and unsupervised and semi-supervised learning in sensory neocortex [71,72]. These forms of learning and adaptation rely in part on temporal and functional flexibility endowed by neuromodulatory processes. NA, for instance, modulates the signal-to-noise ratio of evoked responses in sensory cortical regions [73] and allows for prioritizing information that is relevant to the current behavioral task [11]. ACh can reconfigure functional network connectivity in the neocortex by effectively disabling weak connections from spontaneous circuit activations for a more modular and hierarchical circuit structure biased in favor of feedforward sensory representations over top-down contextual inputs [3].

NA has been traditionally implicated in arousal and attention [74]. The adaptive gain and network reset theories have been proposed to explain the role of NA in behavior [75]. The adaptive gain theory is inspired, in part, by an observed dichotomy in NA neuron activity patterns in relation to behavioral performance. Specifically, these neurons display phasic and tonic activity during optimal and poor behavioral performance, respectively. In the phasic mode, transient increases in NA activity exploit task-specific learning, whereas the tonic activity mode sets in when the task is no longer useful and another exploration starts [74]. The theory, therefore, suggests that NA may optimize the trade-off between exploitation and exploration [74,76], serving as a biological foundation of transitions between strategies [77,78]. The network reset theory, by contrast, hypothesizes that when unexpected cues or relevant contexts mandate behavioral changes, NA neurons are activated to induce arousal and reset network activity [43,79]. These theories, while offering important conceptualizations, likely represent only partial explanations of the role of NA in behavior and cognition. In particular, recent studies have provided mounting evidence for projection-specific functions of NA neurons [80,81] and challenge the view of the NA system as a homogeneous entity, as sometimes depicted in theories about its roles in arousal and attention. Additional domains where NA neuromodulation plays an important role are fear processing and extinction learning. Activation of NA neurons that project to the amygdala facilitates fear learning, whereas overstimulation of NA neurons projecting to the medial prefrontal cortex disrupt retention of extinction memories [82].

The thalamus plays an important role in relaying and gating the flow of sensory information, leading to a refined representation of salient, behaviorally relevant sensory stimuli in the neocortex. It has been argued that tonic spikes arising due to locus coeruleus (LC) activation convey threefold more information compared with tonic spikes in the absence of LC stimulation [83]. Relatedly, LC activation enhances the performance of animals executing a tactile frequency discrimination task through NA regulation of sensory processing in the thalamus [82].

The dual mode of tonic and phasic ACh release in the striatum relates to specific neural pathways: tonic release often arises from local cholinergic interneurons, whereas phasic release stems from projections from the brainstem. In the striatum, the modulatory effects of ACh emerge not only from direct midbrain-striatal projections and local striatal cholinergic interneurons, but also from indirect routing through the thalamus [84]. Cholinergic neurons from the pedunculopontine nucleus and the laterodorsal tegmental nucleus innervate functionally segregated areas of the thalamus (i.e., the relay structures) and the limbic nuclei, respectively [85]. Midbrain cholinergic neurons from both areas innervate the thalamic intralaminar nuclei, which convey sensory information to the striatum. Sensory inputs to the striatum are thought to play important roles in providing behavioral contexts for action selection [86]. These functionally specialized projections may play a role in the transmission and selection of relevant sensory information and encoding of behaviorally relevant stimuli. In the context of DNNs, the distinct connectivity and network topology of the biological ACh system could inspire computations that support processes like avoidance behavior and retention of information in response to negative outcomes, as well as filtering of redundant information. In parallel, modeling the midbrain-thalamo-striatal circuitry in DNNs could lead to a better understanding of the role of striatal ACh in behavioral flexibility, where contradictory findings have been reported [87–89].

A long-standing, yet unresolved debate, in relation to ACh relates to the mode of neuromodulatory release and signaling. Classic descriptions of ACh release in the cortex have centered around the principle of diffuse organization, which has hypothesized that slowly varying concentrations of extracellular ACh could cause state transitions, for instance, between arousal states [90]. The propensity of ACh to diffuse in the extrasynaptic volume (**volume transmission**), in

contrast to being localized to the synaptic cleft (synaptic transmission), has been central to this hypothesis.

Although neuromodulators have often been studied as distinct, independent systems, the widespread presence of neurotransmitter co-release, as well as co-activation of receptors for different neurotransmitters, suggests that complex interactions between neuromodulators could lead to myriad effects [6,91,92]. The physiological relevance of volume transmission has also been discussed considering neuromodulatory systems such as 5-HT and NA [93–95]. Nevertheless, DA, ACh, and NA appear to function through both synaptic and volume transmission [96–98]. Thus, a fundamental feature of neuromodulatory signaling is the ability to operate through distinct modes of release, volume and synaptic or interactions between the two, to bring about a variety of behavioral responses across different timescales. Fast responses could support cognitive functions such as attention or short-term reward processing and slower changes may relate to uncertainty, vigilance, arousal, and long-term reward-related learning [98,99].

Neuromodulation-inspired DNNs

Incorporating principles based on neuromodulation into DNNs could be an important component to enable them to better generalize across tasks and learn over diverse timescales. Learning at fast timescales promotes rapid changes in synaptic weights, which could foster learning with sparse and uncertain data. Slow timescales, however, could allow synapses to remain persistently stable and prevent **catastrophic forgetting** in continual learning tasks [100].

DNNs are widely trained on nonsequential tasks using large data samples that are independent and identically distributed [101], while continual learning without catastrophic forgetting and adaptation to multiple sequential tasks remains challenging [102–105]. We term DNN models that integrate an understanding of global phenomena in the biological brain and local computations across diverse spatio-temporal scales as neuromodulation-aware DNNs. In neuromodulation-aware DNNs, two approaches have been proposed to address this challenge: modular networks and meta-learning. Specifically, through reusing modular components of a DNN for different scenarios and evolving functionally specialized modules, modular continual learning enables new associations when new tasks are introduced [101,106]. Meta-learning is an approach inspired by the ability of brains to generate novel solutions to problems with limited experience [101]. A recent study proposed a neuromodulated meta-learning algorithm to achieve learning without catastrophic forgetting on sequential tasks [107]. In another recent study, prior findings from the *Drosophila* mushroom body network on context-based DA modulation of short-term synaptic plasticity have inspired DNNs that implement efficient learning in one-shot image classification on a small number of samples [108].

Neuromodulators reshape learning, in part, by gating the time window for the induction of synaptic plasticity [109–111]. Neuromodulatory release, associated with the behavioral state of an animal, controls the gating and the directionality of synaptic plasticity across brain regions [110]. In the amygdala, DA suppresses feedforward inhibition and modulates the time window required for long-term changes to enhance synaptic weights [112]. By contrast, in the dorsal striatum, DA governs long-term changes in both strengthening and weakening synaptic connections [113]. In the prefrontal cortex, ACh reverses long-term synaptic potentiation into synaptic depression [114], and in the visual cortex, the concerted effects of ACh and NA result in long-term synaptic potentiation and depression, with NA flipping the depression of synaptic strength to potentiation and ACh performing the opposite [110]. Therefore, mechanistic principles derived from the multiscale functions of neuromodulators in gating synaptic plasticity and transforming

learning rules could inspire a new class of neuromodulation-aware DNNs that continually learn and adapt to ever-changing tasks with minimal supervision and retraining.

The development of brain-inspired DNNs has been enriched by our knowledge of physiological processes, as well as cognitive phenomena at the macroscale [115–117]. An example is the implementation of artificial attention mechanisms (Box 2) [53,104], which have become popular in several domains, including natural language processing [118], computer vision [119,120], and multitask learning [121,122]. More recently, biological details at the dendritic, single-neuron or **microcircuit** levels [123,124] have been increasingly incorporated into DNNs. Neuromodulation-enabled plasticity and auto-reconfiguration of network properties have been implemented in DNN architectures [106,107,125,126] and cognitive robots [127–131]. For example, a recent DNN implementation incorporated neuromodulation for processing contextual information (e.g., information about the current task and environment) by adjusting the slope and bias of the **activation functions** in the DNN [125]. Other studies have modeled the neuromodulation of synaptic plasticity to overcome catastrophic forgetting or to achieve more efficient learning by introducing populations of modulatory neurons and a network-computed neuromodulatory signal, respectively [106,126]. In addition, in some cognitive robot designs, neuromodulated learning has been incorporated in a modular implementation, where the functional roles of 5-HT, Ach, and DA are abstracted at a high behavioral level to enable attention to salient objects, harm aversion, and risk taking in real environments [127,129–131]. In studies using robotic platforms, detection of objects triggers artificial DA- or

Box 2. Biological and artificial attention

In artificial intelligence, biological attention has been approximated through weighting and reweighting processes that generate the relative importance of components of input sequences [53]. Artificial attention, however, is an emerging application-driven simplification of biological attention. Additionally, in artificial attention, weighting in a continual or multitask learning scenario is complicated, as the tasks may yield different objectives that cause the relative importance of the same feature or input component to vary across tasks [53]. In an environment with a larger number of relatively independent tasks, it would be of behavioral relevance to calculate the task-specific relative importance of features [157,158], and a neural network that effectively exploits the relationship and variance across tasks may lead to higher performance.

Biological attention is defined by cognitive functions that facilitate the selection of information relevant for the organism and the filtering of unwanted information. Attention thus enables flexible control and efficient use of limited computational resources in various circumstances [53,145]. In the contexts of neuroscience, cognitive science, and psychology, investigation of attention has been conducted on a broad range of organisms, including rodents, nonhuman primates, and humans, using a wide spectrum of experimental techniques and behavioral tasks (Table I). Experimental studies on neural correlates of attention have revealed biological underpinnings of different attentional forms, for example, visual, auditory, olfactory, and crossmodal attention.

Table I. Experimental techniques used in the investigation of biological attention and their respective temporal resolution, duration, spatial resolution, and target biological organisms

Experimental technique	Temporal resolution	Duration	Spatial resolution	Biological organisms
Single-unit recording	<= ms	Hours, can be up to days	μm	Rodents, non-human primates [146,147]
Electrocorticography (ECoG)	<= ms [148]	Hours, can be up to days	mm to cm	Non-human primates, humans [149,150]
Electroencephalography (EEG)	<= ms	Hours, can be up to days	cm (cortical area to whole brain)	Non-human primates [151], humans
Magnetoencephalography (MEG)	<= ms	Hours	cm (cortical area to whole brain)	Humans [152,153]
Near-infrared spectroscopy (NIRS)	Seconds	Hours, can be up to days	cm (cortical area)	Non-human primates, humans [154,155]
Functional magnetic resonant imaging (fMRI)	Seconds	Seconds to hours	mm to cm (cortical layer to whole brain)	Non-human primates [156], humans

5-HT-like neurons, depending on the level of novelty or danger embodied. ACh and NA were modeled as a filter that gates attention and distills events by adjusting weights from neurons that correspond to sensory events from ACh or NA neurons [130,131].

Thus far, with different objectives in mind, studies have implemented abstract and simplified models of biological neuromodulation, often in the form of mechanisms that dynamically self-reconfigure network activity. Overall, behavioral benefits were observed at multiple spatio-temporal levels and could be attributed to the emergence of divergent intrinsic properties of single modeled neurons and randomly chosen subpopulations of the neural network under varying intensity of neuromodulation [132]. However, future studies, including ones using neural networks with greater biological detail, are required to understand whether neuromodulation-driven transitions of neuronal dynamics yield patterns similar to those observed in a neural network representing a local microcircuit of neurons. Learning in an environment with the goal to maximize long-term reward without access to explicit feedback signals has been studied in the context of reinforcement learning and, more recently, **deep reinforcement learning** [104,133]. Although reinforcement learning has offered explanations of how DA neurons could be recruited as RPE signals [134–137], contributions of other neuromodulatory systems to reinforcement effects on neural circuits in different brain regions, as well as their interactions, are yet to be evaluated [138,139]. In biological brains, neuromodulators reconfigure the functionality of anatomically defined neural networks to generate varied patterns of functional outputs [140]. This can be seen as analogous to task-based resetting or optimization of hyperparameters in DNNs, but performed in a more self-adaptive, unsupervised manner. To add to and complement established DNN architectures [141,142], we propose a framework that integrates parallel neuromodulation at different spatio-temporal specificities through: (i) reconfiguration of network hyperparameters, (ii) cell type-specific neuromodulation, (iii) neuromodulated scaling and update of weights, and (iv) computations using **compartmental model neurons** (Figure 3).

At the network level, hyperparameters including learning rate, dropout probability, and activation functions, could be learnable functions of internal latent representations, environmental state, rewarding or aversive stimuli, level of novelty, etc., to achieve dynamic control of the learning process. In addition, cell type-specific neuromodulation on excitatory and inhibitory cells has been shown to enable inhibition and disinhibition of subpopulations of neurons, making acquisition of opposing behavioral patterns possible [132]. A future step would be to assign neuromodulation in a cell type-specific manner in DNNs to investigate firing activities and learning trajectories with and without neuromodulation. Furthermore, previous attempts have been made to introduce neuromodulated plasticity to DNNs, which improved model performance in spatial learning and language tasks [126]. Integration of similar mechanisms with biologically realistic DNNs could lead to a more comprehensive understanding of whether and how neuromodulated plasticity improves learning efficiency. Finally, one could consider using local, region-specific compartmental neuronal morphologies to enable divergent computations of errors [123], which are in turn dynamically modulated by an input from downstream, global neuromodulatory factors. Combining network-level neuromodulation with environmental cues could also facilitate encoding of the probability of behavioral state transitions, which affect action selection strategies (Figure 3). To understand the learning of neuromodulation-aware DNNs, model performance in both long-term goals and short-term subtasks (e.g., total rewards obtained in a maze task and avoidance of an aversive cue, respectively; Figure 3) should be evaluated.

Conductance-based or compartmental neuron models in computational neuroscience provide more biological realism than point neuron abstractions and are commonly used to simulate the dynamics of single neurons or small networks. However, they typically do not allow for learning

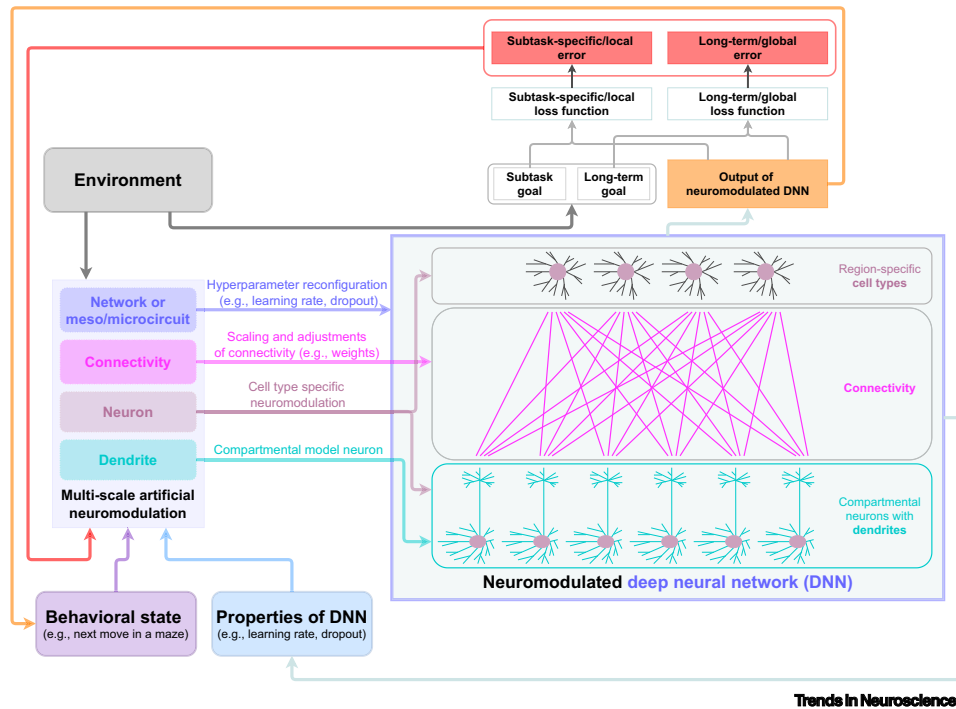


Figure 3. A framework for integrating multiscale neuromodulatory processes into deep neural networks (DNNs). A framework of multiscale artificial neuromodulation in DNNs, with neuromodulatory effects introduced at four spatial scales. A neuromodulation-aware DNN could be implemented using region-specific types of compartmental neuron models. Such a DNN receives updated hyperparameters, while updating its connectivity through dynamic scaling of synaptic weights. At the network scale, network hyperparameters (e.g., learning rate and dropout probability) are regularly updated. In parallel, adjustments of connectivity across neurons are performed through local and/or global scaling of weights, while neuromodulatory factors are applied to different subpopulations of neurons depending on their properties. In addition, biologically detailed neurons may be implemented as the basic computational units of the DNN. The DNN receives information about the environment, as well as its behavioral state and properties. Performance of the DNN is assessed through subtask-specific or long-term loss functions.

in tasks where behavioral responses and adaptations are required [143]. DNNs, by contrast, can achieve high performance in object recognition, language processing, and maze navigation, but are coarse approximations of biological processes. The emerging use of DNNs in computational neuroscience may facilitate convergence between the bottom-up and top-down approaches, which could offer an intuition, for instance, on how neuromodulatory processes may affect neuronal firing in a DNN in response to changes in an environment. Although neuromodulation-inspired DNNs have generated positive outcomes in tasks including maze navigation, future work is necessary to further examine their utility using more complex environments and goals [125,126,132]. In the longer term, neuromodulation-aware DNNs could enable continual and adaptive learning and flexible action selection and planning. Such learning models may allow simulating biological mechanisms that underlie adaptive learning, which are difficult to monitor through experiments alone.

Concluding remarks and future perspectives

The functional flexibility of biological neuromodulatory systems is governed by: (i) spatio-temporal specificities, (ii) reconfiguration of neuronal network dynamics, (iii) differential or synergistic innervations of target structures, and (iv) similarities and divergences across each individual neuromodulatory system. Neuromodulatory processes are a result of dynamic, multiscale

Outstanding questions

The role of DA in reward assessment is well established. DA, however, also interacts with other neuromodulators, including ACh, which contributes to avoidance behavior and motor control. How could neuromodulatory interactions be implemented in DNNs? How could DNNs evaluate the emergence of synergistic behavioral effects?

The NA system comprises neurons with distinct projection patterns, which differentially modulate network-level parameters. Could multiscale implementations of parameter update and projection-specific functions of NA enable DNNs to efficiently reconfigure network-level parameters?

The neuromodulator 5-HT is thought to modulate learning and perception by activating receptor 'hotspots' on the distal apical dendrites of neocortical layer 5 pyramidal cells. Recently, DNNs have used dendritic mechanisms to address the credit assignment problem. Would incorporating dendritic receptor clusters in DNNs help unravel the role of neuromodulators in gating sensory perception?

Neuromodulators are thought to simultaneously reorganize neural connections and enable synaptic weight updates, resulting in network state shifts, for example, (in biological networks) from distraction to attention. How could DNNs exploit simultaneous mechanisms of rewiring and weight updates to bring about state transitions in changing environments?

Neuromodulators such as ACh and NA promote overall renormalization and homeostasis of synaptic strength during different states of sleep, and facilitate learning and memory through replay. Replay enables the brain to consolidate memory and overcome forgetting of acquired knowledge, also referred to as 'catastrophic forgetting' in the computational literature. How could DNNs incorporate 'sleep-like states' to mimic biological replay mechanisms and prevent catastrophic forgetting?

competition and cooperation of neuromodulators, regulating both local, transient events and global, persistent cognitive and physiological phenomena across different brain regions. The spatio-temporal precision of neuromodulatory processes may enable a global reconfiguration of neuronal networks through modification of local synaptic connections. Biological neuromodulation also acts to integrate neural events that occur within a timescale of milliseconds and connects neural events of various timescales to behavioral signals of longer delays and durations. Therefore, neuromodulatory systems could provide insights into how brains implement reinforcement learning and behavioral flexibility.

Despite rich evidence for the roles of neuromodulators in governing learning and adaptive behavior, it remains challenging to build neuromodulation-aware DNNs that reflect the properties of brain cognition, which are shaped by its various components (see [Outstanding questions](#)). Therefore, it may be important for DNNs to identify the level of biological details required to achieve more adaptive, generalizable learning. To enable the next generation of neuromodulation-aware DNNs capable of continual and adaptive learning, we propose a complementary multiscale framework that allows implementation of neuromodulatory mechanisms across multiple scales: hyperparameters, connectivity, and single units. This framework promotes mapping learning trajectories of DNNs by evaluating model performance in both short-term subtasks (e.g., identification of changes in an environment for replanning the path) and long-term goals (e.g., navigation to a location). Neuromodulation-aware DNNs could permit a deeper understanding of the causal relationship between neuromodulatory processes and learning outcomes in functional settings.

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Declaration of interests

The authors declare no competing interests in relation to this work.

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